

Internship Domain: Machine Learning

Arch Technologies - Month 2 Project

Submitted By:

Full Name: Yalda Ashrafi

Email Address: yaldaashrafi54@gmail.com

Phone Number: +601113304425

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Introduction

This project focuses on building a machine learning model to predict housing prices using the California housing dataset. By leveraging supervised learning techniques, the model analyzes key features such as location, number of rooms, and population to automate price estimation. The project encompasses the entire machine learning life-cycle: data cleaning, feature engineering, model training, and performance evaluation.

Importing Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error, r2_score

df = pd.read_csv("housing.csv")
```

View the column names

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20640 entries, 0 to 20639
Data columns (total 15 columns):
```

#	Column	Non-Null Count	Dtype
0	longitude	20640 non-null	float64
1	latitude	20640 non-null	float64
2	housing_median_age	20640 non-null	float64
3	total_rooms	20640 non-null	float64
4	total_bedrooms	20640 non-null	float64
5	population	20640 non-null	float64
6	households	20640 non-null	float64
7	median_income	20640 non-null	float64
8	median_house_value	20640 non-null	float64
9	ocean_proximity_INLAND	20640 non-null	bool
10	ocean_proximity_ISLAND	20640 non-null	bool
11	ocean_proximity_NEAR BAY	20640 non-null	bool
12	ocean_proximity_NEAR OCEAN	20640 non-null	bool
13	rooms_per_household	20640 non-null	float64
14	pop_per_household	20640 non-null	float64

dtypes: bool(4), float64(11)
memory usage: 1.8 MB

View the first 5 rows

```
df.head()

{"summary": "{\n  \"name\": \"df\",\n  \"rows\": 20640,\n  \"fields\": [\n
```

Data Cleaning

```
imputer = SimpleImputer(strategy="median")
df['total_bedrooms'] = imputer.fit_transform(df[['total_bedrooms']])
df = pd.get_dummies(df, columns=['ocean_proximity'], drop_first=True)
```

Feature Engineering

```
df['rooms_per_household'] = df['total_rooms'] / df['households']
df['pop_per_household'] = df['population'] / df['households']
```

Model Training

```
X = df.drop("median_house_value", axis=1)
y = df["median_house_value"]
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

model = RandomForestRegressor(n_estimators=100, random_state=42)
model.fit(X_train_scaled, y_train)

RandomForestRegressor(random_state=42)
```

Evaluation

```
y_pred = model.predict(X_test_scaled)
print(f"R2 Score: {r2_score(y_test, y_pred):.4f}")
print(f"RMSE: {np.sqrt(mean_squared_error(y_test, y_pred)):.2f}")

R2 Score: 0.8108
RMSE: 49,793.89
```

Residual Plot

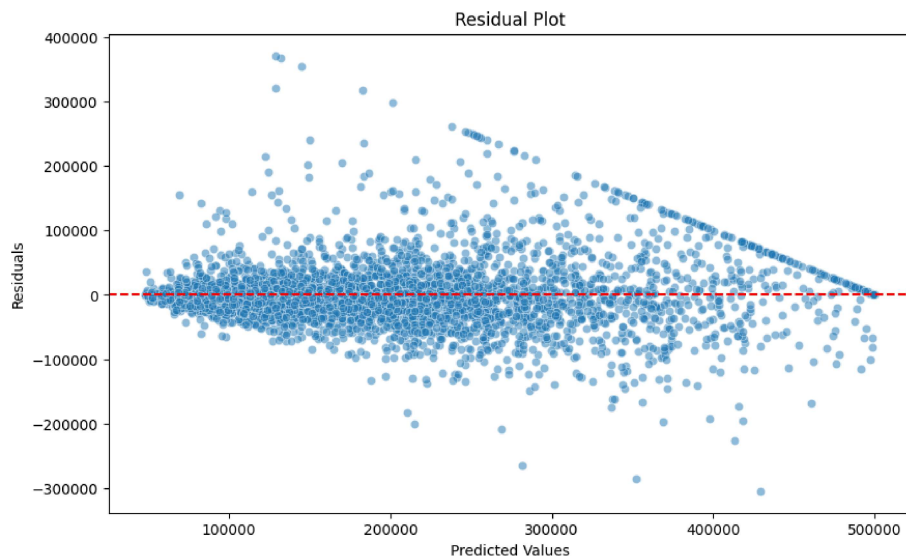
A residual plot helps to visualize the errors of a prediction model. It displays the residuals (the difference between observed and predicted values) on the y-axis and the independent variable (or predicted values) on the x-axis. A good model will have residuals randomly scattered around the horizontal line at zero, with no discernible pattern. If a pattern is observed, it indicates that the model might not be capturing some underlying structure in the data.

Model Diagnostic: Residual Plot Analysis

```
y_pred = model.predict(X_test_scaled)
residuals = y_test - y_pred

plt.figure(figsize=(10, 6))
sns.scatterplot(x=y_pred, y=residuals, alpha=0.5)
plt.axhline(y=0, color='r', linestyle='--')
```

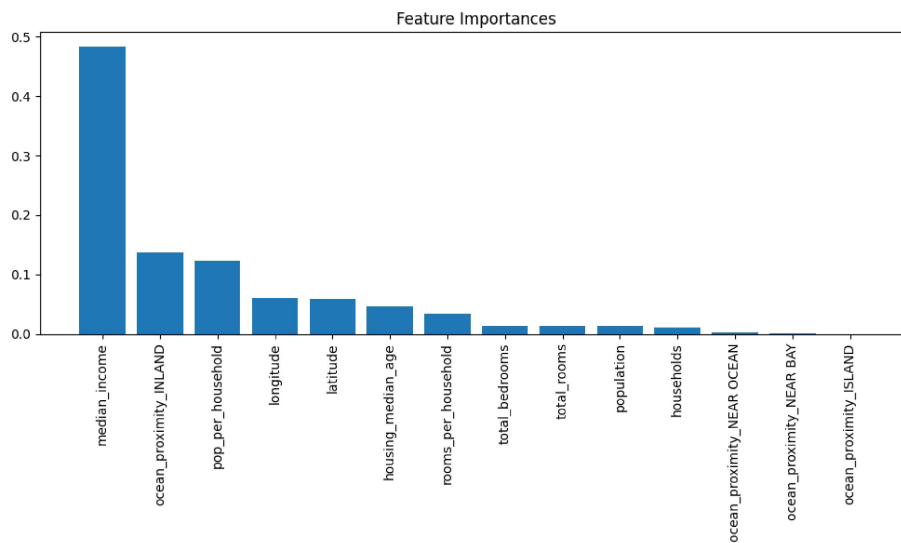
```
plt.title('Residual Plot')
plt.xlabel('Predicted Values')
plt.ylabel('Residuals')
plt.savefig("residual_plot.png")
plt.show()
```



Feature Importance Bar Chart

```
importances = model.feature_importances_
indices = np.argsort(importances)[::-1]
feature_names = X.columns

plt.figure(figsize=(10, 6))
plt.title("Feature Importances")
plt.bar(range(X.shape[1]), importances[indices], align="center")
plt.xticks(range(X.shape[1]), [feature_names[i] for i in indices], rotation=90)
plt.tight_layout()
plt.savefig("feature_importance.png") # Saves to your Colab files
plt.show()
```



Project Methodology Summary

1.Data Preprocessing & CleaningMissing

Data: The total_bedrooms column contained missing values, which were addressed using a Median Imputer. Categorical Encoding: The ocean_proximity feature was transformed using One-Hot Encoding to convert categorical geographic data into a numerical format suitable for machine learning algorithms.

2. Feature Engineering

To enhance model performance, new features were created, including rooms_per_household and pop_per_household, providing the model with more descriptive insights into housing density and demographics.

3. Model TrainingAlgorithm:

A Random Forest Regressor was selected for its robustness in handling non-linear relationships. Scaling: Features were processed using StandardScaler to ensure consistency in data ranges, allowing for more accurate model convergence.

4. Model Evaluation Metrics:

The model's accuracy was assessed using the R^2 Score (measuring explained variance) and the RMSE (measuring prediction error in dollars). Diagnostics: Visualizations, including a Correlation Heatmap, Feature Importance Bar Chart, and Residual Plot, were generated to interpret the model's logic and validate prediction reliability

R^2 Score (0.8108):

This is the coefficient of determination. It tells us that our model successfully captures 81.08% of the variability in housing prices based on the provided features. In machine learning, a score closer to 1.0 indicates a high-performing model that explains the data well.

RMSE (49,793.89):

This is the Root Mean Squared Error. It represents the average prediction error in dollars. It signifies that, on average, our model's predicted house price is off by approximately 49,794 compared to the actual market value.

Project Repository:

<https://github.com/Yalda-Ashrafi/Arch-Technologies-ML-Internship-Month2.git>